

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

BIOMOLECULES

LECTURE-7

ZOOLOGY

DAVOOD SIR (CEO of NG)

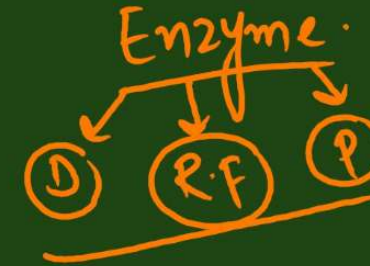


Physics Wallah

Topics To Be Covered

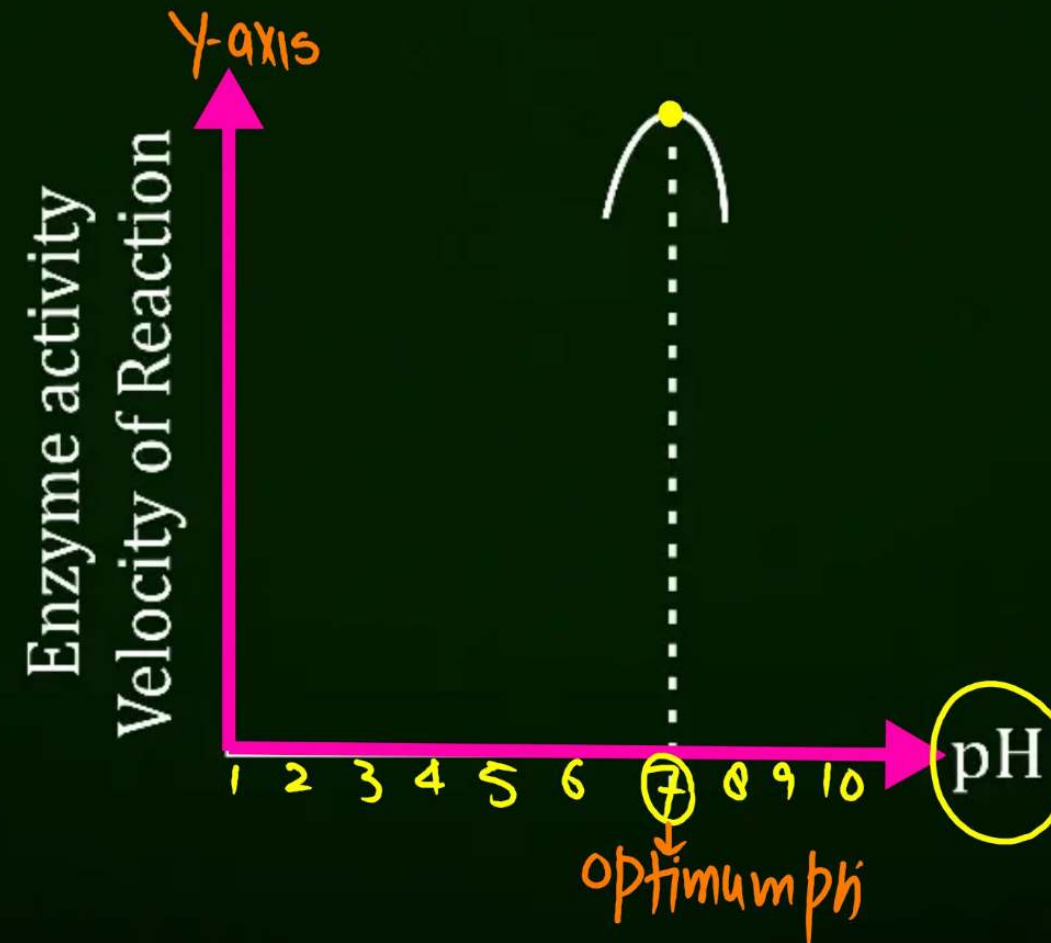
1 **Enzymes** → Part-2.

2 **Previous Year NEET MCQ's**



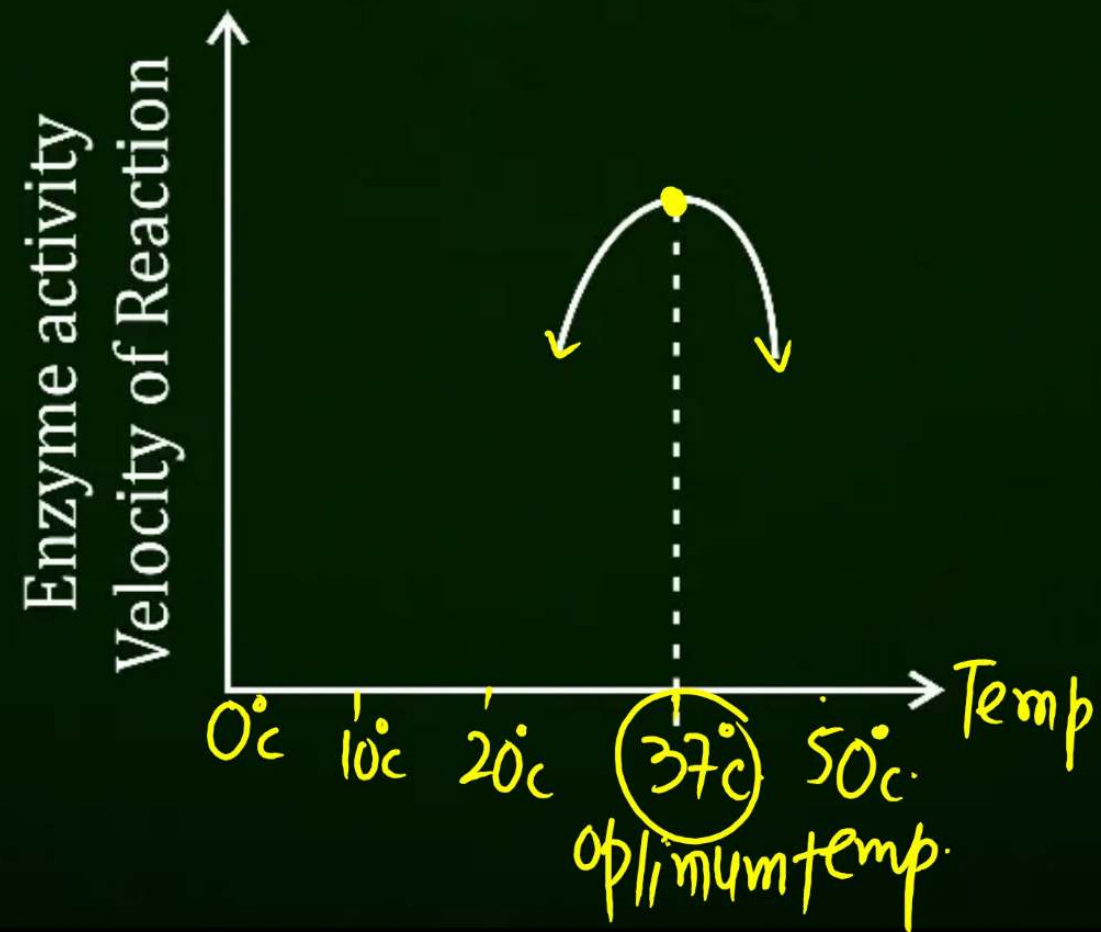
Factor Effecting Enzymes Activities

1. pH: Enzymes function within a narrow pH range, showing maximum activity at a specific value known as the optimum pH. Activity decreases if the pH moves above or below this point. Drastic pH changes can alter the enzyme's shape (denaturation), making it ineffective.



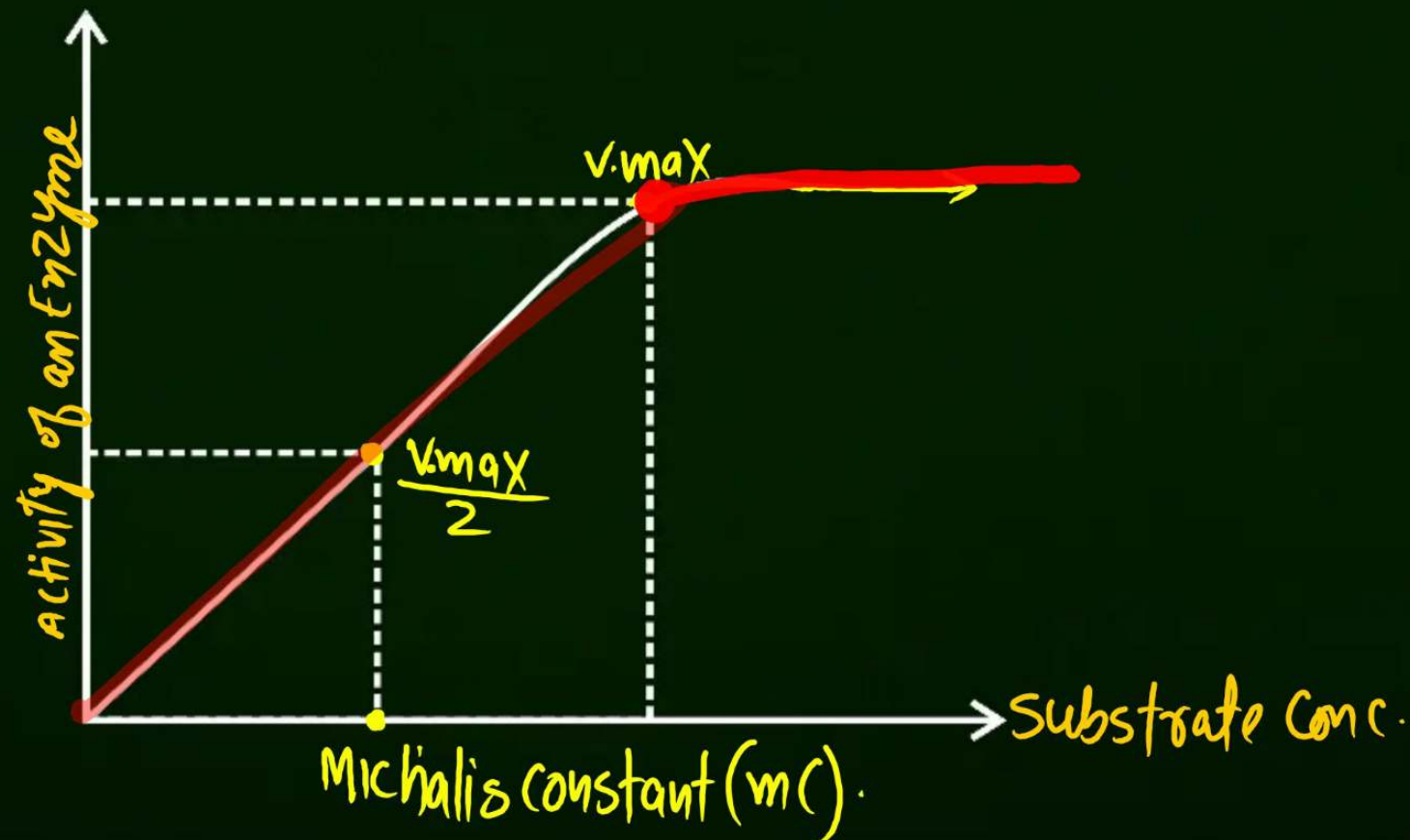
Factor Effecting Enzymes Activities

2. Temperature: Enzymes function within a narrow temperature range, with peak activity at an optimum temperature. Activity drops if the temperature is too low or too high. Low temperatures make enzymes temporarily inactive, while high temperatures denature the proteins, permanently destroying their activity.



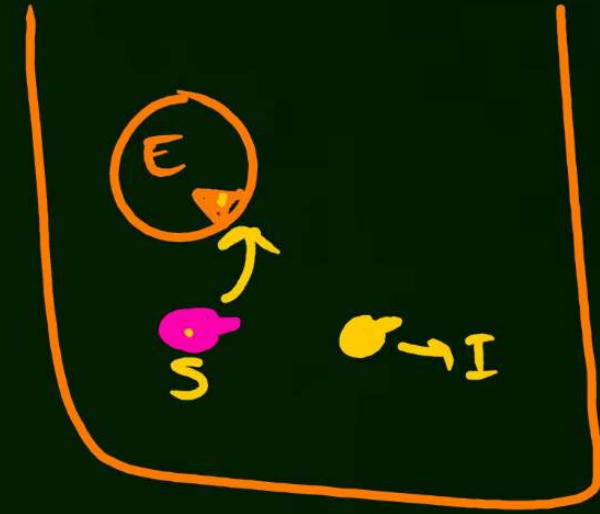
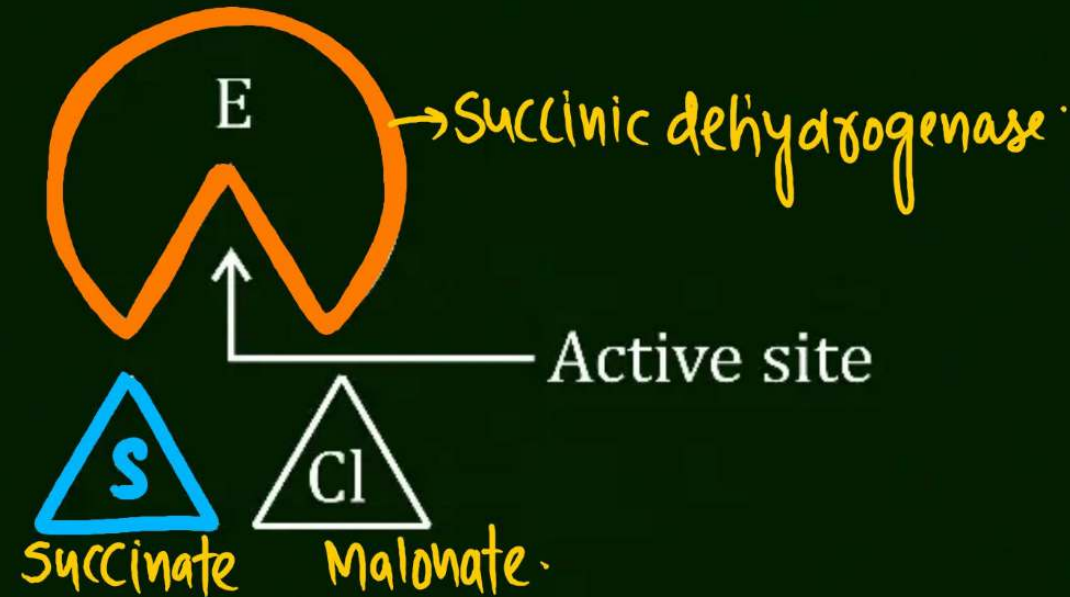
Factor Effecting Enzymes Activities

3. Concentration of Substrate: As substrate concentration increases, the rate of the enzymatic reaction initially rises. Eventually, it reaches a maximum velocity (V_{max}) beyond which no further increase in rate occurs. This is because all enzyme molecules become saturated no free enzymes are left to bind with excess substrate.



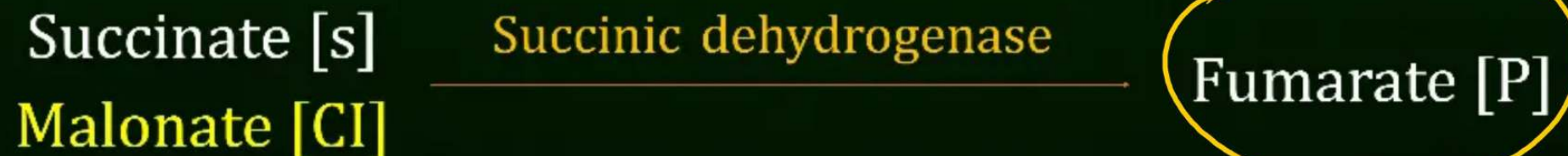
Factor Effecting Enzymes Activities

4. Inhibitors:- When the binding of any chemical shuts off enzyme activity, the process is called inhibition and the chemical is called an inhibitor.



❖ Types of Inhibitors:-

1. Competitive Inhibitors (CI):- They are structurally similar to the Substrate & bind to Active site of Enzyme



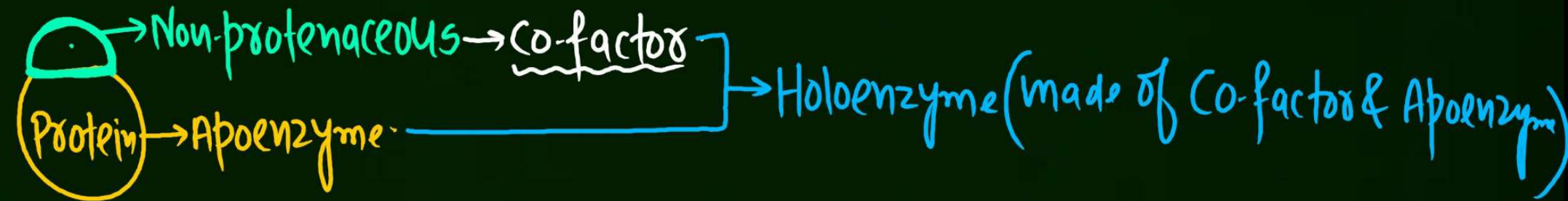
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How does Temperature and pH affect the enzyme Activity?

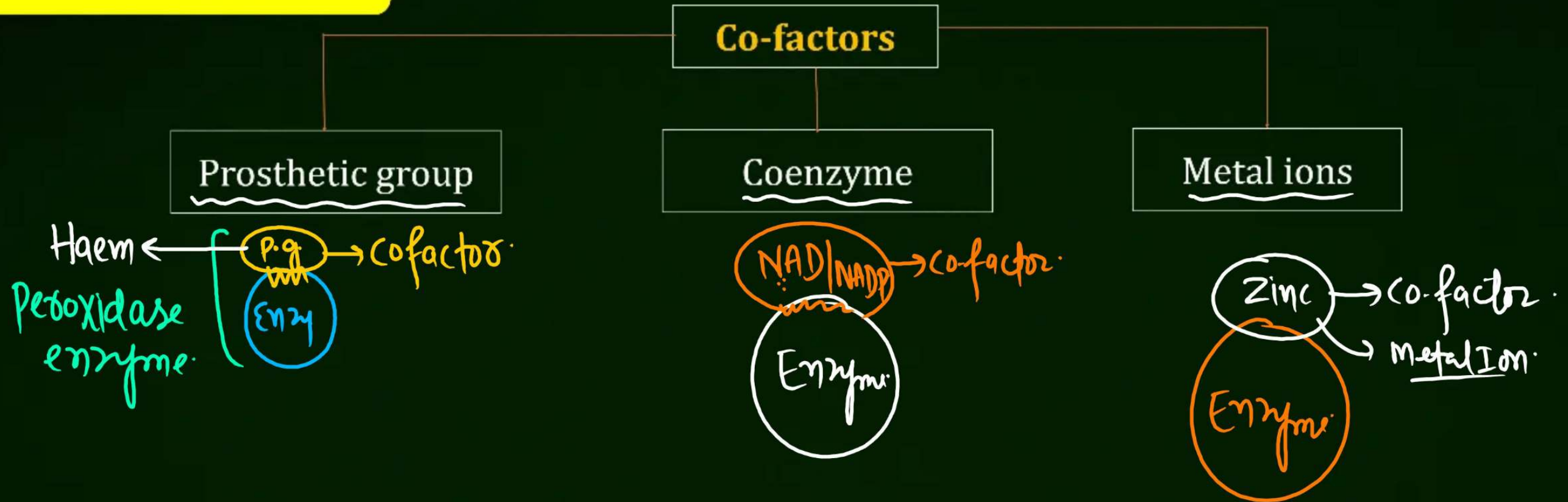


Co-Factors

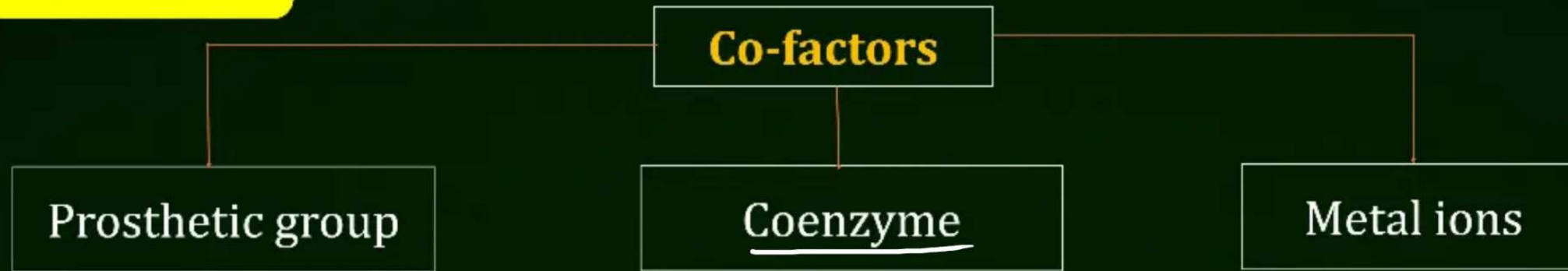
- Some enzymes need non-protein part along with protein part to become catalytically active.
- The protein part alone is called the **apoenzyme** and non-protein part is called **Co-Factor** & both are called **Holoenzyme**.
- Catalytic activity stops if the cofactor is removed.



Co-Factors



Co-Factors



- **Prosthetic Groups:**
 - Organic compounds, tightly bound to apoenzyme.
 - Example: Haem in peroxidase and catalase (breaks H_2O_2 to water and oxygen).
- **Co-enzymes:**
 - Organic compounds, loosely and temporarily bound.
 - Often derived from vitamins (e.g., NAD and NADP from niacin).
- **Metal Ions:**
 - Required by some enzymes; form coordination bonds with enzyme and substrate.
 - Example: Zinc in carboxypeptidase.

↓
v. B₃

Classification of Enzymes

➤ According to Enzyme Commission (EC), Enzymes are classified into 6 major classes each with 4-13 subclasses based on the type of reaction they catalyse.

✓ 1. Dehydrogenases/Oxido-Reductases:- Enzymes which catalyze oxidation-reduction between two substrates. Example: *Alcohol dehydrogenase (ADH)*.



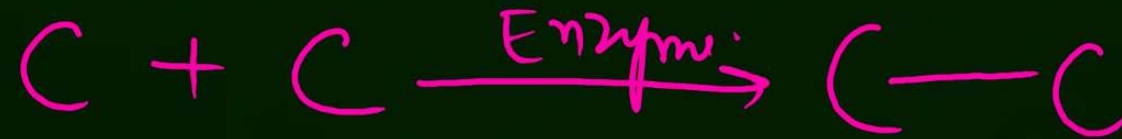
✓ 2. Transferases:- Enzymes which transfer groups from one molecule to another. Example: e.g. *Kinase*.



✓ 3. Hydrolases:- Enzymes that break down large molecules into smaller ones by using water in a chemical reaction. E.g:- lipases

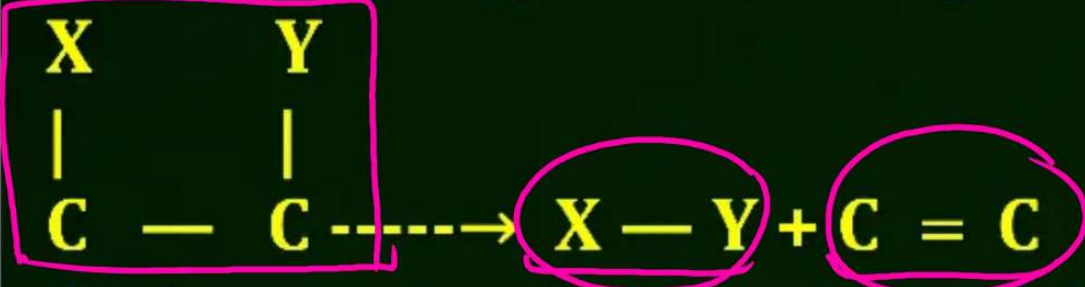
Classification of Enzymes

- ✓ 4. **Lyases:-** Enzymes that catalyze the breaking of chemical bonds by means other than hydrolysis. *E.g. Carbonic anhydrase*.
5. **Isomerases:-** Enzymes that catalyze the structural rearrangement of molecules, converting one isomer into another. *E.g. Glucose isomerase*
6. **Ligases :** Enzymes that join two molecules by forming a new covalent bond. They play a key role in DNA replication, repair, and protein synthesis. *E.g. DNA ligase*



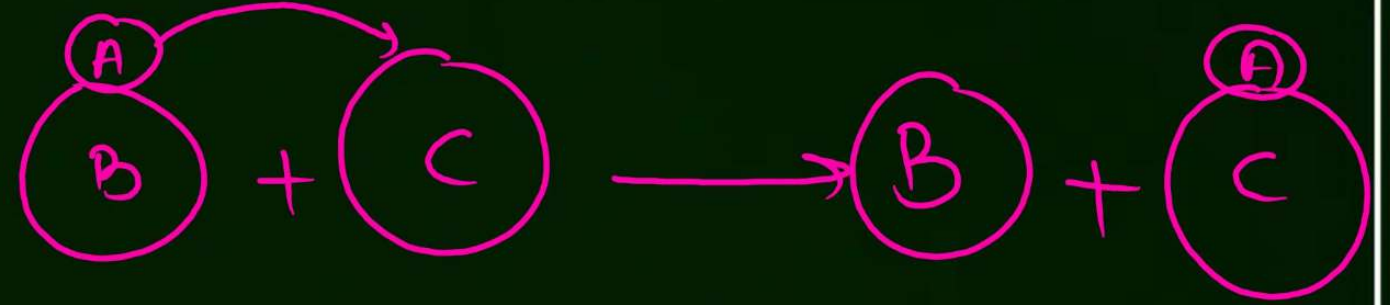
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The following reaction depicts the activity of a particular class of enzymes:



(Substrate) (Product) (Product)

Identify the enzyme class 'E' from the following options:



A

Ligases ~~X~~

B

Lyases ✓

C

Isomerases ~~X~~

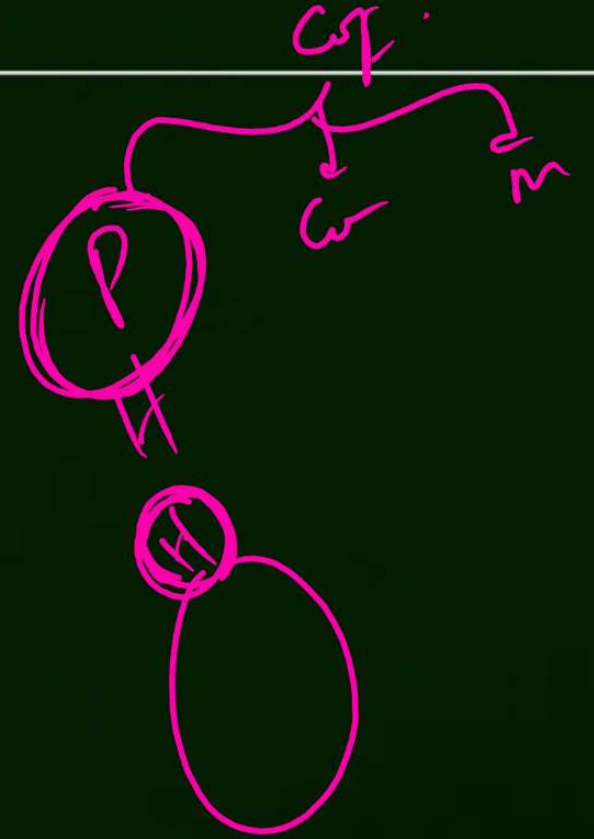
D

Transferases ~~X~~

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Which one of the following enzymes contains 'Haem' as the prosthetic group?

- A** Carbonic anhydrase
- B** Succinate dehydrogenase
- C** Catalase
- D** RuBisCo



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Name the class of enzyme that usually catalyze ~~of~~ the following reaction:



Where, G → a group other than hydrogen

S → a substrate

S[#] → another substrate

A Lyase

B Transferase

C Ligase

D Hydrolase



Thank
You